

COMP 8005

Assignment 2

Report

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Purpose

This program will attempt to crack a password that is hashed using one of the following algorithms: MD5, SHA-256, SHA-512, bcrypt, and yescript, and multithreading in the worker.

Requirements

Task	Status
Crack the password and display results	Fully implemented

Platforms

controller.py and worker.py has been tested on:

- Ubuntu 2023.10

Language

- Python

Documents

- [Design](#)
- [Testing](#)

Findings

Design

Overall Architecture

The architecture of this program is a multithreaded controller/worker architecture, and uses heartbeat protocols

Controller/Worker Responsibilities

Controller

The controller is responsible for setting up a server, and to parse out the shadow file information from the given user. The controller will parse out the: user, hash algorithm, salts, and the hashed password. This information will be passed onto the worker that connects with it to crack the password. The controller will also send and receive heartbeat requests/responses to verify the workers status.

Worker

The worker is responsible for connecting with the server/controller and waiting for a job. After receiving the password information, the worker will then create a given number of threads to use the hash algorithm and password information to use a brute-force method to crack the password, and will return the results back to the controller. The worker will also receive heartbeat requests and give a response with the number of attempts tried.

Communication Protocol

The controller and worker communicate through a TCP socket connection. The controller will start the TCP socket, and await a connection. The worker will then connect, and await for a response from the server where it will receive the cracking information. After cracking the password, the worker finishes by sending a final response with all the necessary information to the controller.

The controller/worker will also use a heartbeat protocol where the controller will send a request to the worker to verify the worker is both working and to check its status.

Partitioning Strategy

The worker will split the legal charset into the number of threads inputted, splitting up the work evenly.

Implementation

Languages and Libraries

Languages

This program was created using Python 3.13.

Libraries

socket: Create the TCP connection

pickle: Dump the communication information for the connection

argparse: Parse command line arguments
datetime: Calculates run time lengths
ipaddress: Check if an IP address is valid
bcrypt: Hash algorithm for bcrypt
crypt: Hash algorithm for yescrypt, SHA-256, SHA-512, MD5
itertools: Iterates through a list of characters
string: Creates list of characters to iterate through
threading: Handles multithreading and locks

Experimental Results

Algorithm Run Times

All runs were done using the following password: abc, A2#, 456

1 Thread

MD5	Min (seconds)	Max (seconds)	Mean (seconds)
Controller Parsing Time	0.0000490	0.0000570	0.0000543
Job Dispatch Latency	0.0001000	0.0001290	0.0001183
Cracking Time	0.5940020	33.0475480	16.4942393
Return Latency	0.0001520	1.1366620	0.4639550
Total end-to-end Runtime	0.5943370	34.1843960	16.9583670

bcrypt	Min (seconds)	Max (seconds)	Mean (seconds)
Controller Parsing Time	0.0000600	0.0000820	0.0000743
Job Dispatch Latency	0.0000760	0.0001140	0.0000983
Cracking Time	9.3247150	528.0375230	263.9936773
Return Latency	0.0522200	1.4309290	0.7201443
Total end-to-end Runtime	10.0021730	528.0899290	264.7139943

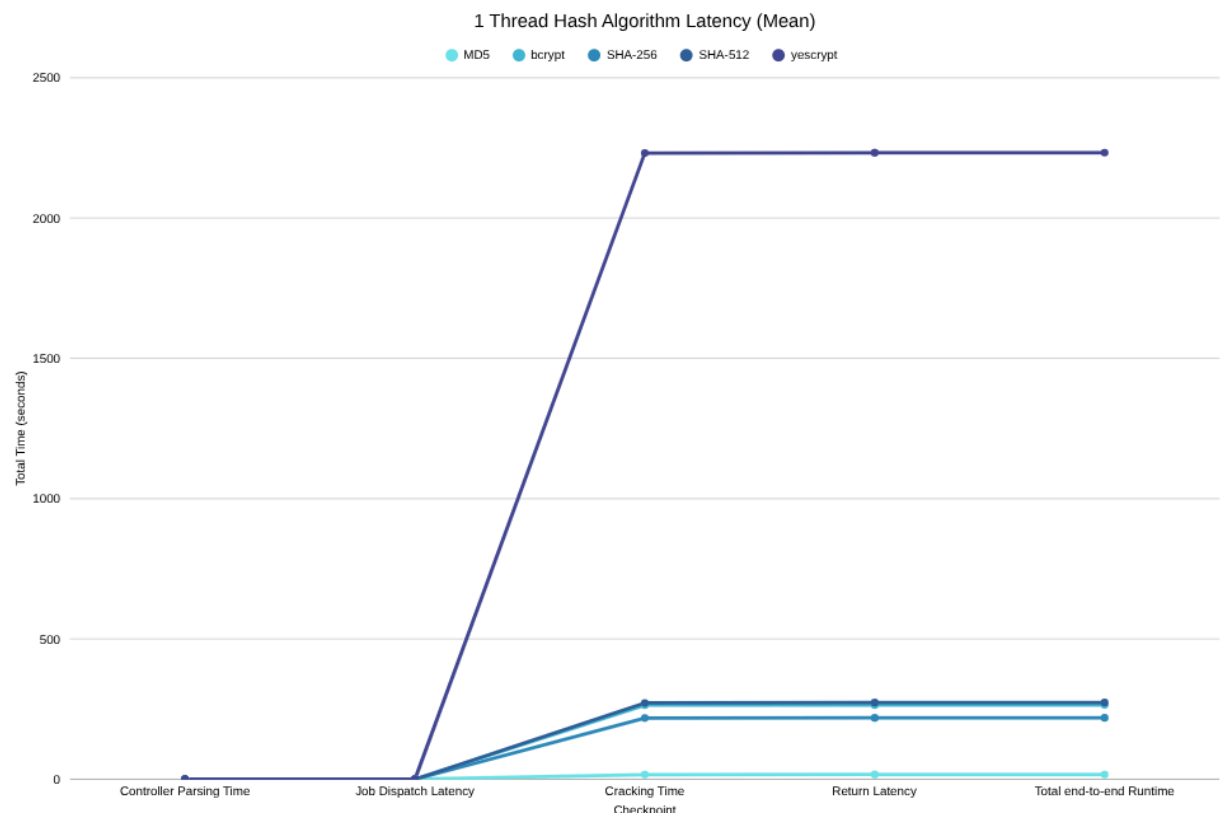
SHA-256	Min (seconds)	Max (seconds)	Mean (seconds)
Controller Parsing Time	0.0000580	0.0000630	0.0000610
Job Dispatch Latency	0.0000880	0.0001420	0.0001117

Cracking Time	7.5503850	437.9172990	218.1651727
Return Latency	0.4336150	1.1545130	0.6901173
Total end-to-end Runtime	8.0327550	439.0720170	218.8554627

SHA-512	Min (seconds)	Max (seconds)	Mean (seconds)
Controller Parsing Time	0.0000550	0.0000650	0.0000610
Job Dispatch Latency	0.0000920	0.0001480	0.0001230
Cracking Time	9.5840080	544.1483300	271.9724240
Return Latency	0.4788170	1.8843880	1.3116193
Total end-to-end Runtime	10.0630380	545.7201380	273.2842273

yescrypt	Min (seconds)	Max (seconds)	Mean (seconds)
Controller Parsing Time	0.0000580	0.0000690	0.0000647
Job Dispatch Latency	0.0000820	0.0001460	0.0001210
Cracking Time	78.0221830	4,461.9833920	2,231.4369410
Return Latency	0.7519420	1.8246320	1.2106887
Total end-to-end Runtime	78.7743270	4,463.8081750	2,232.6478153

Graph



2 Threads

MD5	Min (seconds)	Max (seconds)	Mean (seconds)
Controller Parsing Time	0.0000530	0.0000580	0.0000550
Job Dispatch Latency	0.0000790	0.0001080	0.0000900
Cracking Time	0.6025570	31.1223310	17.5492490
Return Latency	1.0852880	1.3979240	1.2488060
Total end-to-end Runtime	2.0006140	32.2077550	18.7982000

bcrypt	Min (seconds)	Max (seconds)	Mean (seconds)
Controller Parsing Time	0.0000530	0.0000580	0.0000550
Job Dispatch Latency	0.0000790	0.0001080	0.0000900
Cracking Time	0.6025570	31.1223310	17.5492490
Return Latency	1.0852880	1.3979240	1.2488060
Total end-to-end Runtime	2.0006140	32.2077550	18.7982000

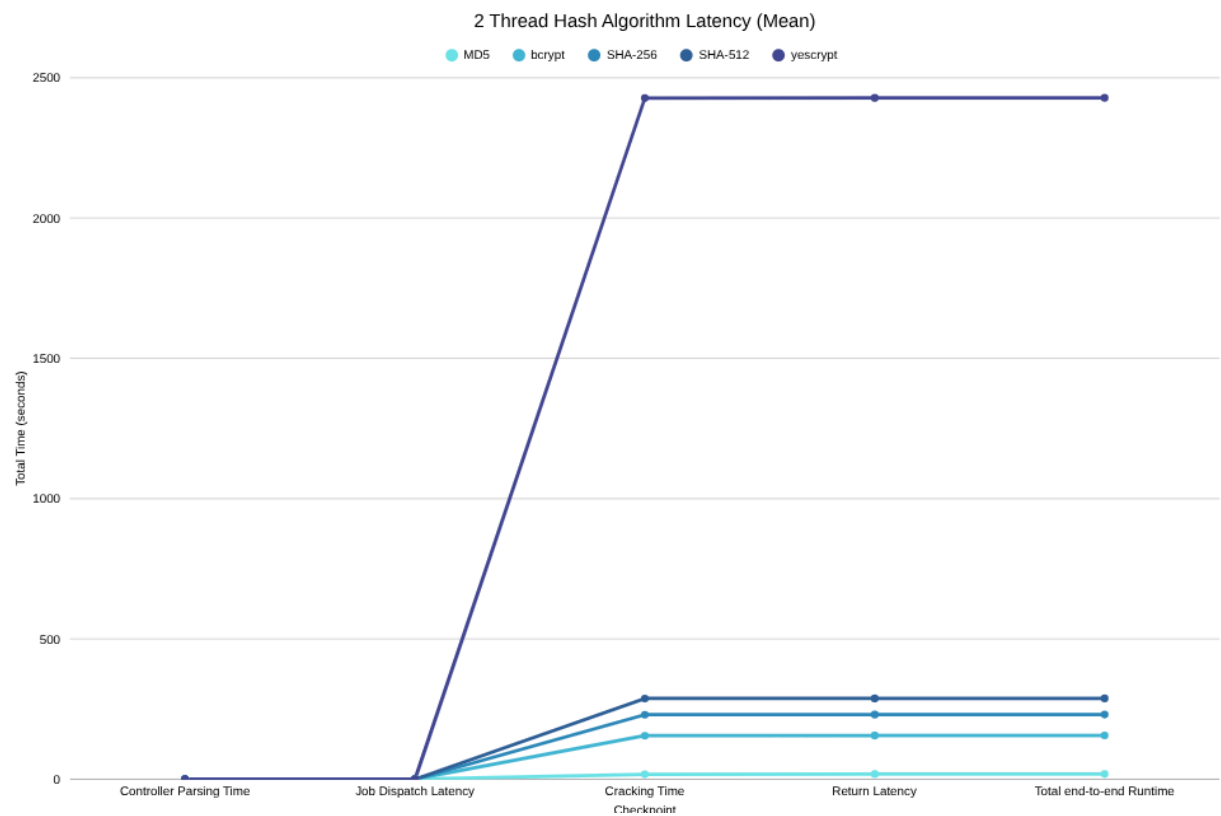
Controller Parsing Time	0.0000510	0.0000800	0.0000610
Job Dispatch Latency	0.0000930	0.0001070	0.0001017
Cracking Time	5.0341960	275.8393350	155.3263730
Return Latency	0.2033770	0.9669040	0.6983747
Total end-to-end Runtime	6.0012590	276.0428560	156.0249103

SHA-256	Min (seconds)	Max (seconds)	Mean (seconds)
Controller Parsing Time	0.0000510	0.0000600	0.0000567
Job Dispatch Latency	0.0001030	0.0001320	0.0001157
Cracking Time	7.5599950	409.0518160	230.0306370
Return Latency	0.4353920	0.7550050	0.5660097
Total end-to-end Runtime	8.0677810	409.4873990	230.5968190

SHA-512	Min (seconds)	Max (seconds)	Mean (seconds)
Controller Parsing Time	0.0000570	0.0000780	0.0000660
Job Dispatch Latency	0.0001100	0.0001460	0.0001273
Cracking Time	10.0130690	515.5434170	287.8604020
Return Latency	0.0442070	0.4232020	0.1872560
Total end-to-end Runtime	10.0575000	515.6379490	288.0478513

yescrypt	Min (seconds)	Max (seconds)	Mean (seconds)
Controller Parsing Time	0.0000600	0.0000670	0.0000637
Job Dispatch Latency	0.0000830	0.0001280	0.0001113
Cracking Time	77.7121610	4,317.2751770	2,427.2652040
Return Latency	0.1547920	1.9870490	0.7760803
Total end-to-end Runtime	77.8671450	4,317.4617600	2,428.0414593

Graph



3 Threads

MD5	Min (seconds)	Max (seconds)	Mean (seconds)
Controller Parsing Time	0.0000500	0.0000530	0.0000517
Job Dispatch Latency	0.0000960	0.0001270	0.0001077
Cracking Time	1.2069080	9.7688960	4.4184253
Return Latency	0.2970750	1.7570860	0.9492733
Total end-to-end Runtime	2.0007440	10.0661190	5.3678580

bcrypt	Min (seconds)	Max (seconds)	Mean (seconds)
Controller Parsing Time	0.0000500	0.0000530	0.0000517
Job Dispatch Latency	0.0000960	0.0001270	0.0001077
Cracking Time	1.2069080	9.7688960	4.4184253
Return Latency	0.2970750	1.7570860	0.9492733
Total end-to-end Runtime	2.0007440	10.0661190	5.3678580

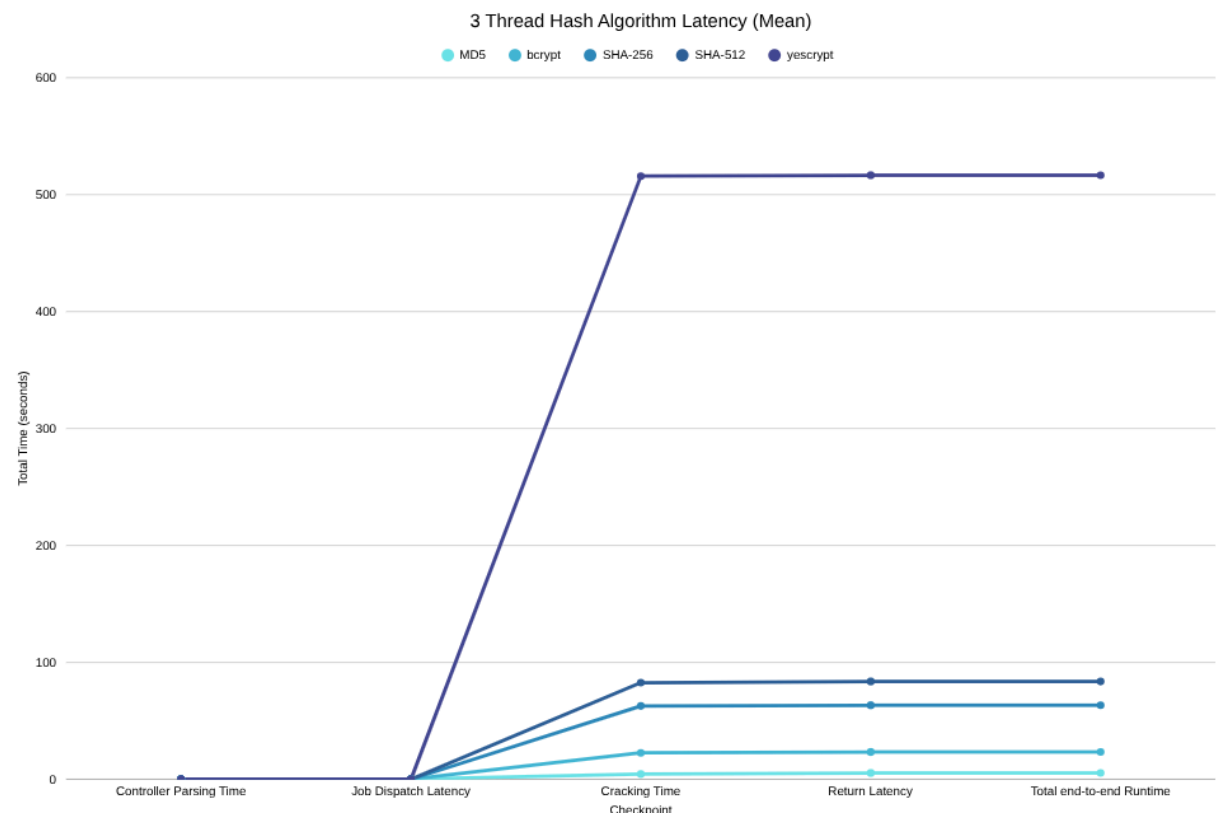
Controller Parsing Time	0.0000570	0.0000580	0.0000573
Job Dispatch Latency	0.0001000	0.0001190	0.0001090
Cracking Time	3.5434420	53.6124580	22.5583230
Return Latency	0.3951260	1.4830100	0.7785260
Total end-to-end Runtime	4.0010490	54.0077410	23.3370153

SHA-256	Min (seconds)	Max (seconds)	Mean (seconds)
Controller Parsing Time	0.0000510	0.0001110	0.0000720
Job Dispatch Latency	0.0000970	0.0001330	0.0001090
Cracking Time	10.1206910	144.5324090	62.6604363
Return Latency	0.0001950	1.0213540	0.6588130
Total end-to-end Runtime	10.1210700	145.4875070	63.3194303

SHA-512	Min (seconds)	Max (seconds)	Mean (seconds)
Controller Parsing Time	0.0000520	0.0000640	0.0000583
Job Dispatch Latency	0.0000860	0.0001120	0.0001003
Cracking Time	13.9137960	196.6057610	82.5822460
Return Latency	0.3185750	1.6741830	1.0618613
Total end-to-end Runtime	14.2325260	198.2801150	83.6442660

yescrypt	Min (seconds)	Max (seconds)	Mean (seconds)
Controller Parsing Time	0.0000580	0.0000800	0.0000657
Job Dispatch Latency	0.0000870	0.0000910	0.0000887
Cracking Time	80.4927620	1,223.9324530	515.8353430
Return Latency	0.0088760	1.8303090	0.7430200
Total end-to-end Runtime	80.5017850	1,225.7629110	516.5785173

Graph



4 Threads

MD5	Min (seconds)	Max (seconds)	Mean (seconds)
Controller Parsing Time	0.0000520	0.0000640	0.0000570
Job Dispatch Latency	0.0001040	0.0001350	0.0001213
Cracking Time	0.5689790	53.9665630	25.8388973
Return Latency	0.8400410	1.4315030	1.2069253
Total end-to-end Runtime	2.0006590	54.8067720	27.0460010

bcrypt	Min (seconds)	Max (seconds)	Mean (seconds)
Controller Parsing Time	0.0000520	0.0000640	0.0000570
Job Dispatch Latency	0.0001040	0.0001350	0.0001213
Cracking Time	0.5689790	53.9665630	25.8388973
Return Latency	0.8400410	1.4315030	1.2069253
Total end-to-end Runtime	2.0006590	54.8067720	27.0460010

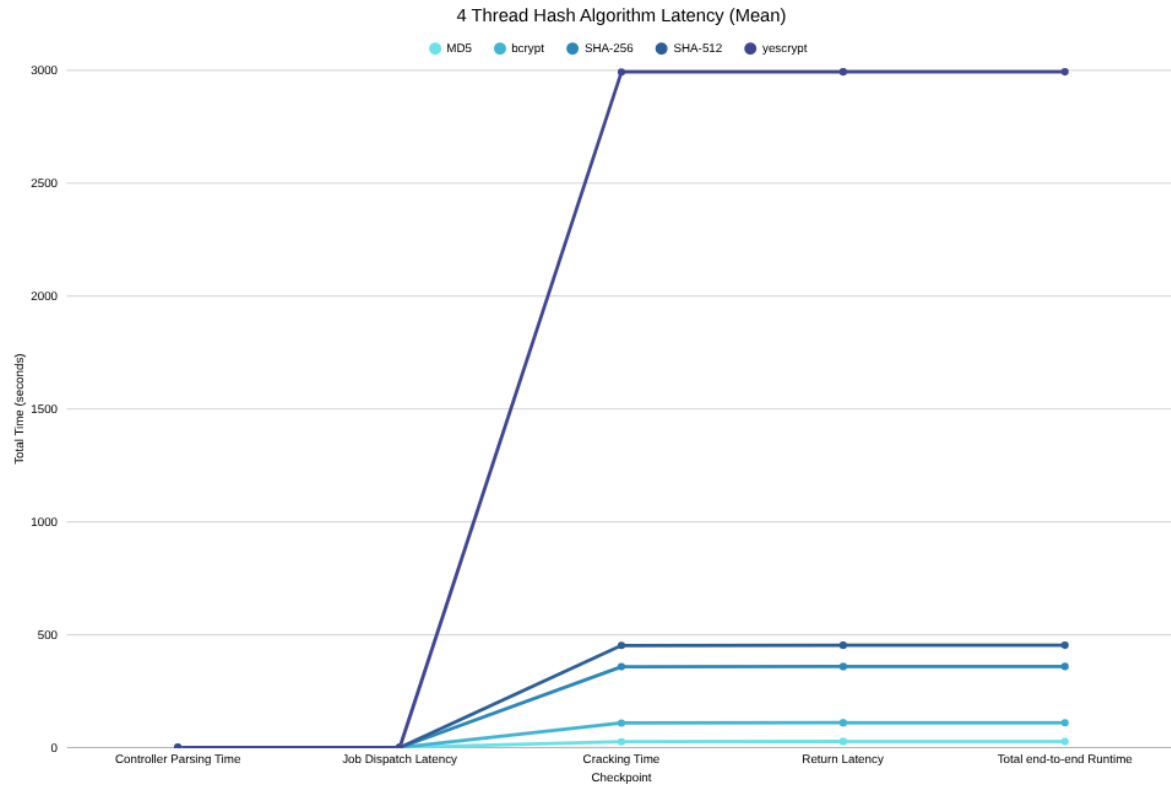
Controller Parsing Time	0.0000560	0.0000800	0.0000643
Job Dispatch Latency	0.0000890	0.0001340	0.0001133
Cracking Time	2.6268040	231.3166130	108.7911640
Return Latency	0.7142410	1.5816560	1.2232110
Total end-to-end Runtime	4.0006850	232.0310450	110.0145527

SHA-256	Min (seconds)	Max (seconds)	Mean (seconds)
Controller Parsing Time	0.0000510	0.0000760	0.0000600
Job Dispatch Latency	0.0001000	0.0001350	0.0001230
Cracking Time	10.3688730	761.1838950	358.2430273
Return Latency	0.1340410	1.8313200	0.8529637
Total end-to-end Runtime	12.2003440	761.7776130	359.0961740

SHA-512	Min (seconds)	Max (seconds)	Mean (seconds)
Controller Parsing Time	0.0000590	0.0000650	0.0000623
Job Dispatch Latency	0.0000900	0.0001130	0.0000987
Cracking Time	14.3896700	941.4661400	452.3245320
Return Latency	0.2938870	1.8387620	1.1330907
Total end-to-end Runtime	16.2286040	942.7329160	453.4577837

yescrypt	Min (seconds)	Max (seconds)	Mean (seconds)
Controller Parsing Time	0.0000550	0.0000590	0.0000570
Job Dispatch Latency	0.0001040	0.0001230	0.0001147
Cracking Time	81.7188490	6,313.3258830	2,992.2168213
Return Latency	0.3166710	1.5709880	0.9228107
Total end-to-end Runtime	82.0356940	6,314.2068340	2,993.1398037

Graph



10 Thread Prediction

I predict the 10 threads will make the program only slightly more efficient, as the number of threads that can be handled is most likely not enough for 10 threads to concurrently run.

10 Threads

MD5	Min (seconds)	Max (seconds)	Mean (seconds)
Controller Parsing Time	0.0000510	0.0000560	0.0000530
Job Dispatch Latency	0.0001010	0.0001510	0.0001273
Cracking Time	0.3102100	78.1087850	29.2946493

Return Latency	0.7747570	1.7215310	1.3955433
Total end-to-end Runtime	2.0007040	79.8304980	30.6903730

bcrypt	Min (seconds)	Max (seconds)	Mean (seconds)
Controller Parsing Time	0.0000520	0.0000780	0.0000647
Job Dispatch Latency	0.0000850	0.0001380	0.0001090
Cracking Time	1.4532640	12.5878840	5.1759070
Return Latency	0.0001710	1.4145110	0.6539437
Total end-to-end Runtime	1.4869600	14.0025510	5.8300243

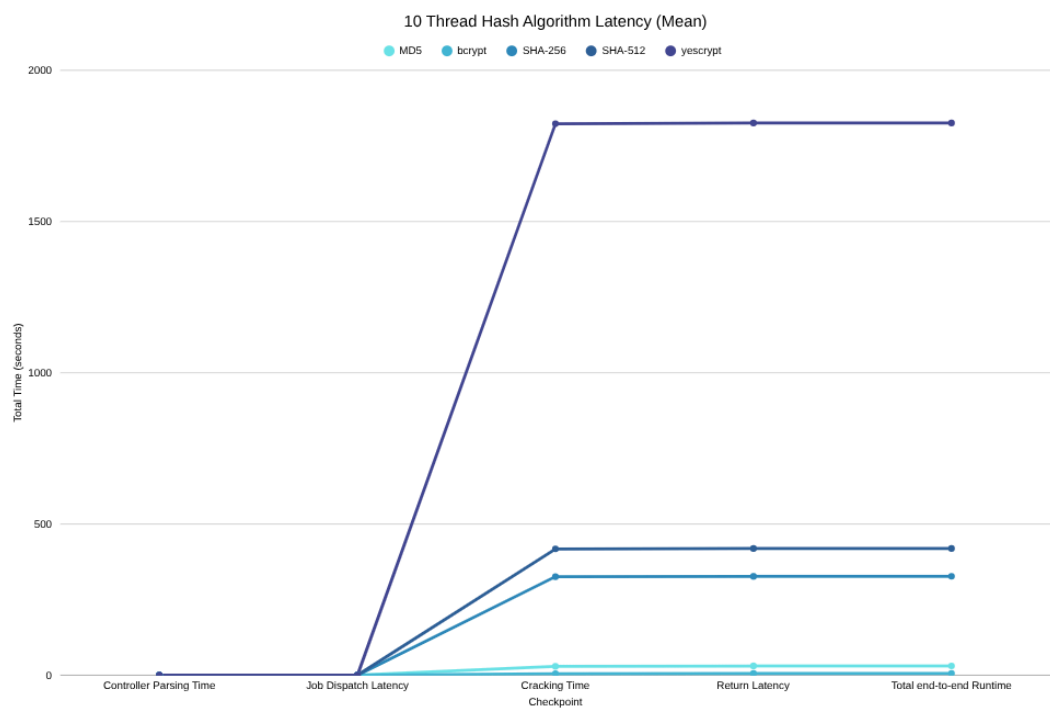
SHA-256	Min (seconds)	Max (seconds)	Mean (seconds)
Controller Parsing Time	0.0000520	0.0000570	0.0000540
Job Dispatch Latency	0.0000960	0.0001170	0.0001037
Cracking Time	16.3559550	838.1170160	325.6865687
Return Latency	0.7771490	1.9174980	1.2080383
Total end-to-end Runtime	17.2855760	840.0346640	326.8947647

SHA-512	Min (seconds)	Max (seconds)	Mean (seconds)
Controller Parsing Time	0.0000530	0.0000630	0.0000567
Job Dispatch Latency	0.0000890	0.0001360	0.0001160
Cracking Time	16.3123980	1,078.3123190	417.5741003
Return Latency	0.6646710	1.8440110	1.4310170
Total end-to-end Runtime	16.9772590	1,080.1564720	419.0052900

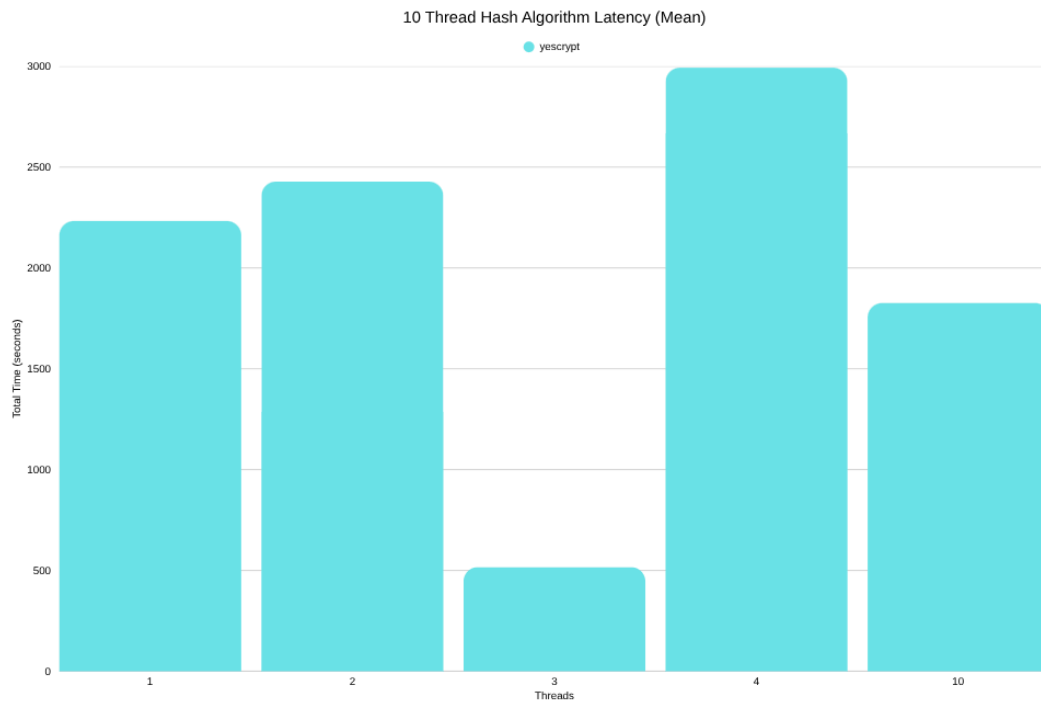
yescrypt	Min (seconds)	Max (seconds)	Mean (seconds)
Controller Parsing Time	0.0000530	0.0000970	0.0000723
Job Dispatch Latency	0.0000870	0.0001260	0.0001077

Cracking Time	85.8692540	4,714.4718630	1,823.0343493
Return Latency	0.4374350	6.2311340	2.6388830
Total end-to-end Runtime	87.1174970	4,720.7031510	1,825.6734123

Graph



Yescrypt Thread Difference Graph



Analysis

For every algorithm, the bottlenecks were the cracking time. The latency between sending jobs or returning data, or even parsing the shadow file are miniscule compared to the time it takes to crack the password. In some cases, multithreading sped up the process, but not consistent enough. This most likely has to do with my implementation of partitioning my legal character set between the threads. Some passwords would show up early in the partition when split into 3 threads, but not when split into 4 threads, causing the 3 thread tests to be substantially faster than the other tests.